

Activity 0.1.3 Choosing the Best Solution

PLTW

CHOOSING THE BEST SOLUTION



Making Decisions and Communicating Ideas



Concept Sketching



Industrial Designers



Conclusion

Making Decisions and Communicating Ideas

GOALS	MATERIALS	RESOURCES
• Generate a decision matrix to assist with idea selection. • Review and practice concept sketching.		
GOALS	MATERIALS	RESOURCES
• Plain paper • Graph paper • Ruler • Pencil		

GOALS	MATERIALS	RESOURCES
• <u>Decision Matrix</u> • <u>Concept Sketching</u> • <u>Concept Sketching Rubric</u> • <u>Activity 0.1.3 Choosing the Best Solution (Downloadable PDF)</u>		

Imagine it's a hot summer day. The best way to cool off might be to head to the local ice cream shop.

What flavor do you get? How about the size? Any toppings? Cup or cone? Or maybe a sundae or milkshake! With so many options, how would you decide?

Many factors can go into your decision. Maybe you are holding your younger sibling's hand, so you choose a cone over a cup. Or it might be an hour before dinner, and so you opt for a small, rather than a medium or large.

These are example criteria that play into your decision-making process. Most of these considerations are likely second nature and do not require a tremendous amount of thought. When engineers make design decisions, they go through a similar process, although it is a bit more structured or intentional.



Another consideration engineers face is how best to communicate ideas. In the early stages of the design process, a sketch is an effective tool to showcase different features of a design. In this activity, you are challenged to think like an engineer to make informed decisions and effectively communicate ideas through sketches.

Decision Matrix

At this stage of your shoe design process, you have developed several potential solutions to your client's problem. These solutions likely excel in some areas, but might have shortcomings in other areas. One solution might be a great option in most categories, but is too costly. In this case, an engineer must consider a trade-off. To make decisions as objectively or quantifiable as possible, engineers use a decision matrix.



Engineering Notebook

To document your process, include your decision matrix and concept sketches in your PLTW Engineering Notebook.

1

Use the [Decision Matrix](#) resource to learn more about this tool and download an example template.

2

Generate appropriate criteria for your shoe designs and complete a decision matrix to choose the best solution.

Concept Sketching

A finalized decision matrix helps you to select the best possible solution for your client. Engineers also develop concept sketches, similar to that in Figure 1, to effectively communicate a design to team members, contractors, and clients.

While some people have greater innate sketching ability than others, you can develop skills through practice, and following a few, simple guidelines.



Figure 1. Water Bottle Concept Sketch

HISTORICAL HIGHLIGHT

Pictures for Communication

Pictures are one of the earliest forms of written communication. Tens of thousands of years ago, ancestral humans depicted scenes on stone, which we commonly call “cave paintings (Figure 2).” These [pictographs](#) were created by mixing pigments from plants or soils with a liquid and applying them to rock faces.

More commonly found are [petroglyphs](#) (Figure 3), scenes carved into rock, which therefore, do not fade as quickly as paintings. Both were used to depict a scene or capture an instant in time, much like a social media post does today!

Later in human history, people began stringing together symbols to tell a story (Figure 4).

[Cuneiform](#) and [hieroglyphics](#) are both early forms of written language. Cuneiform, developed by the Sumerians of Mesopotamia, uses a system of wedge-shaped characters. Hieroglyphics use pictures that represent words or sounds.



Figure 2. Pictograph



Figure 3. Petroglyph

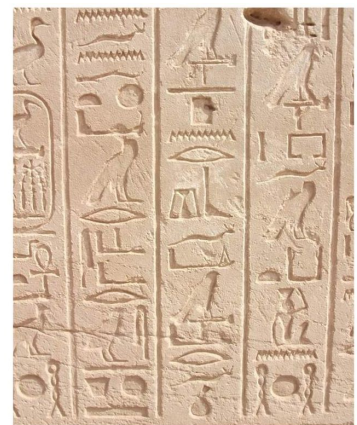


Figure 4. Hieroglyphs



4

To practice, sketch a common object around you.

5

Exchange your work with a partner and provide written feedback to each other using the **Concept Sketching Rubric**.

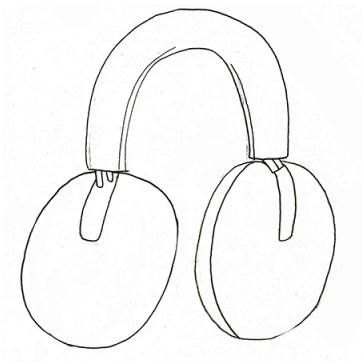
6

Repeat steps 4 and 5 at least two more times. Challenge yourself to incorporate your partner's feedback to improve with each subsequent sketch.

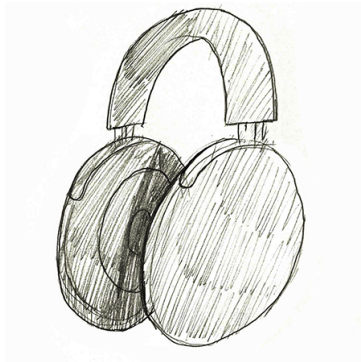
Reflection



The three concept sketches in the flashcards below each attempt to convey a design solution for a pair of headphones. Look at each one, identify any shortcomings, and determine which sketch best conveys the product. When ready, check your thinking by selecting each image in turn.



Lack of shading and one ear muff facing the wrong way.



Shading added (only two tones).
Ear muff orientation fixed.



Shading spectrum indicates
light source, material texture,
and contours.

Now that you have some sketching practice under your belt, create a concept sketch of your shoe design. In this sketch, highlight notable features you want to convey to your audience using [callout](#) labeling as in Figure 5 or describe these features in a list.

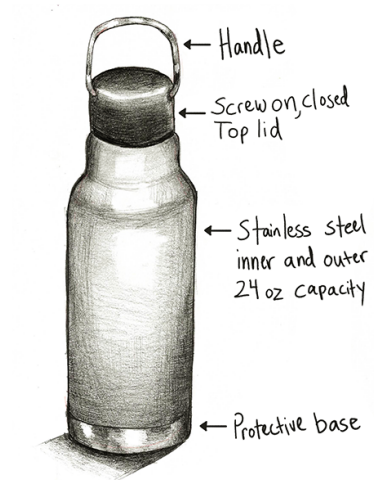


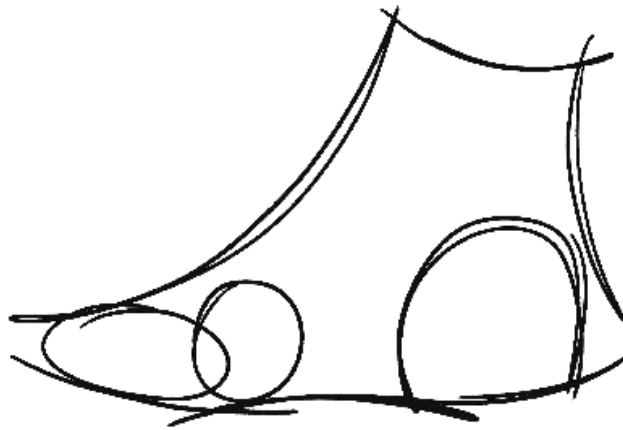
Figure 5. Sample Concept Sketch with Callouts

NOTE: If you have trouble with your sketch, you can use the Sketching a Shoe resource that walks you through sketching a generic shoe.

Sketching a Shoe

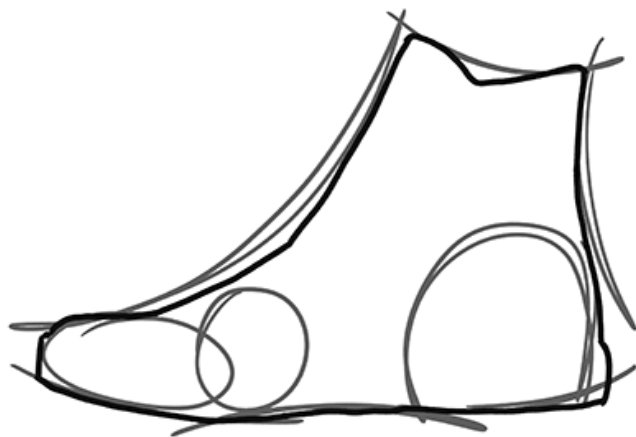
Learn more about...

Step 1



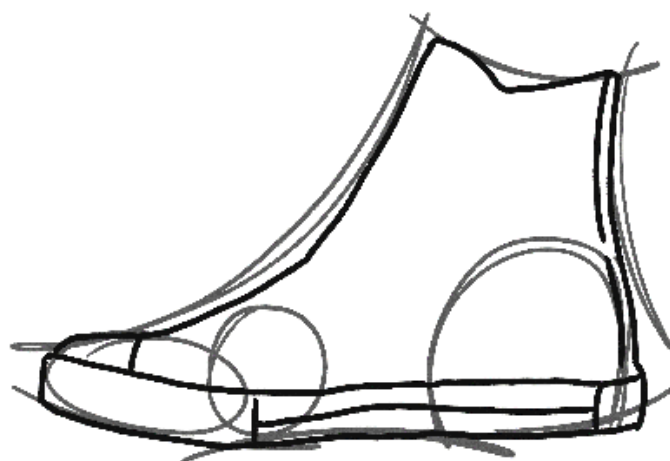
Start with basic shapes to make an outline.

Step 2



Refine your outline.

Step 3



Add details, like the sole.

Step 4



Add more details and texture.

Step 5



Erase guidelines.

Step 6



Add shading to provide texture and depth.

Industrial Designers



CAREER CONNECTION

Mark Guarraia | *Industrial Designer*

Growing up, Mark enjoyed science and math but was passionate about art. He relished opportunities to incorporate creativity in school assignments. At a young age, Mark had an eye for iteration and improvement.

He remembers feverishly sketching a re-designed passenger area of the family minivan during long road trips! Mark's fascination with cars did not stop there. He would obsess over their styling, performance, and competitive characteristics. This fueled his interest in product design, which he saw as the perfect marriage of form and function.



It took Mark's high school art teacher to help point out that this creative passion could ultimately serve well in a field called industrial design. He received a Bachelor of Fine Arts (BFA) from the Rhode Island School of Design (RISD), where he was exposed to novel materials, viewpoints, and methodologies. Studio courses included human-powered vehicle exploration, furniture design, product invention labs, and CAD. While the courses and faculty were great, Mark was most grateful for the diverse and immensely talented network of students he met while at RISD.

“As a student, allow yourself to be wrong so that you can learn from those mistakes. Too often, I see students trying to be perfect and afraid of failure. Invariably, their creativity is limited. If there is ever a time to take a risk and be “wrong,” it is while in school!”

Shortly after graduating, Mark began doing industrial design work in the healthcare industry. He enjoyed several years of hands-on design work on a wide range of projects, from conceptualizing very aspirational and explorative research devices to designing low-cost and robust drug delivery devices for countries that lack sufficient access to care.

For many people in technical careers, there comes a pivotal moment where you must decide either to continue honing your craft and become a specialist, or transition into leadership. In Mark’s experience, choosing the latter proved difficult at first, because he was forced to let go of the hands-on design work. Over time, he began to enjoy using his experience to shepherd young designers, helping them improve their skills.

Mark is currently Vice President of Design and User Experience at Novo Nordisk, one of the largest pharmaceutical companies in the world. Based in Copenhagen, Denmark, the company is focused on tackling some of the most challenging chronic illnesses in the world. Mark’s team strives to remove the burden of illnesses from patients by safely and effectively delivering medication. This means designing the physical and digital devices that millions of patients rely on daily.

Mark feels fortunate to have spent the majority of his career within healthcare. Every project he has influenced has been conceived and executed with the intention to help those in need.

As someone who has been in the field for a while, Mark has some words of wisdom to pass on:

- Worry less, experiment more.
- Seek out professional mentors early, even while still in high school.
- Never shy away from approaching a company you want to work for; perhaps they have not posted for interns or entry-level talent, but the best companies respond well to dedicated and driven talent who can clearly articulate the passion and value they bring to the table.
- Be competitive with yourself, first and foremost.
- Remember to stay humble, and remind yourself there will always be someone better than you at a particular skill; rather than feel defeated by this, use it to fuel your own development and passion.



Reflection



- What resonates most with Mark's story?
- Regarding his words of wisdom, what are you already doing well?
- What could you work on that might have the most impact on your future success?

Conclusion

Question 1

Describe the benefits of using a decision matrix over a less intentional, less structured process.

Question 2

Explain four ways to produce sketches that accurately capture details of your intended design.